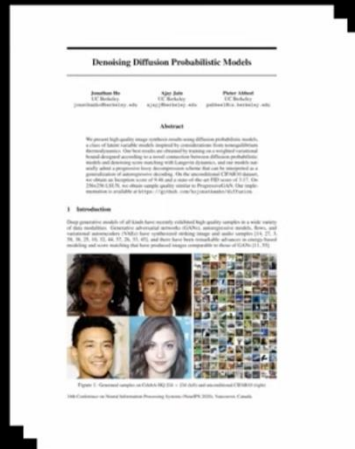


Diffusion Models

Denoising Diffusion Probabilistic Models

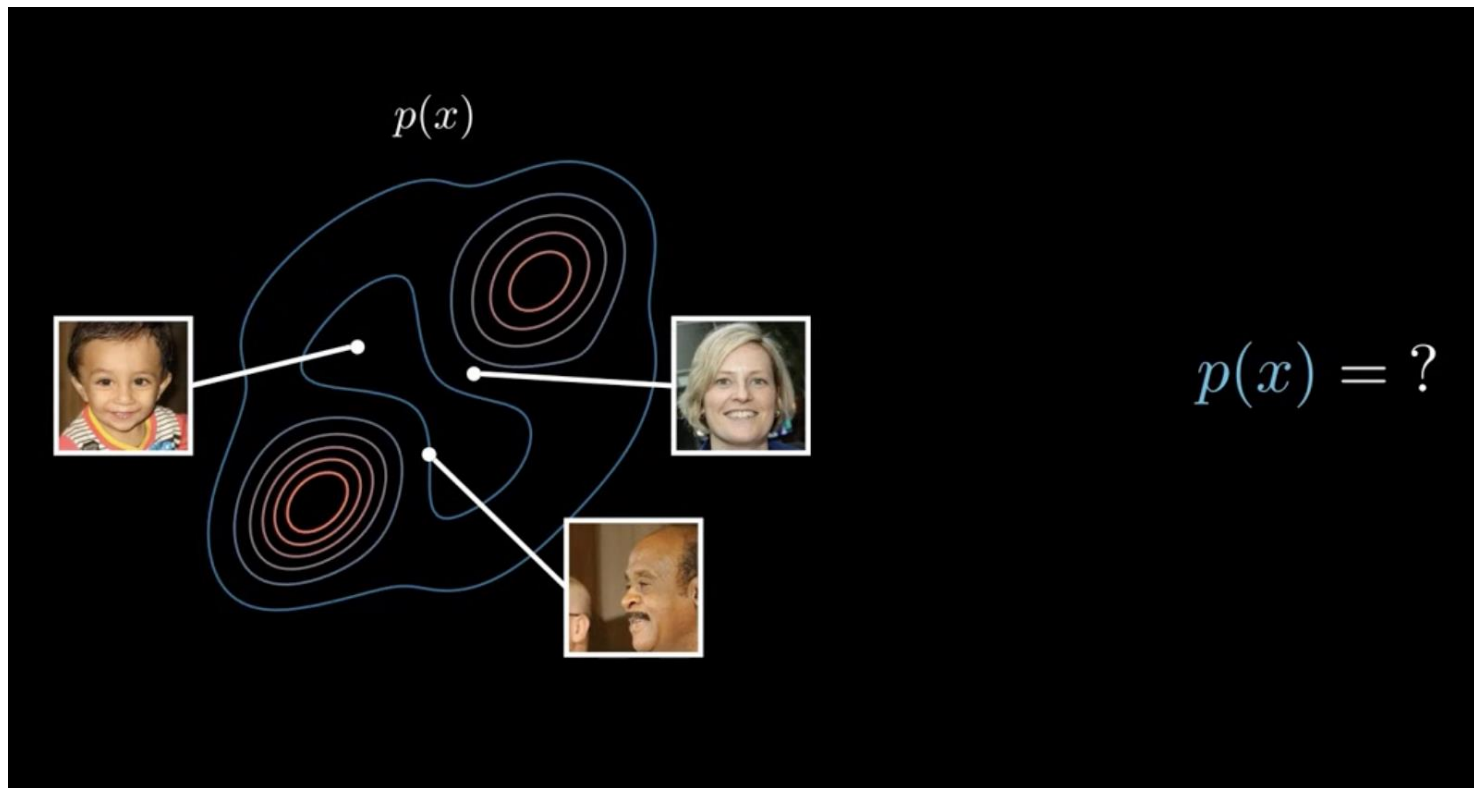
Denoising Diffusion Probabilistic Models

2020



THE PROBLEM

We do not know $p(x)$



Diffusion-based

Forward diffusion process (fixed)



X_0

X_1

X_2

X_3

X_4

...

X_T



Reverse diffusion process (must learn)



Forward Diffusion Process



x_0

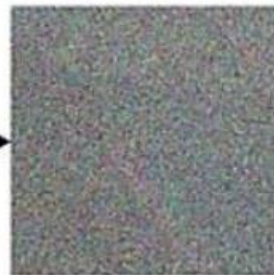


x_1



x_2

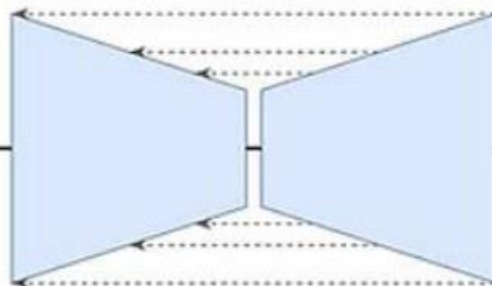
...



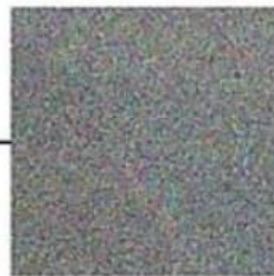
x_T



x_0



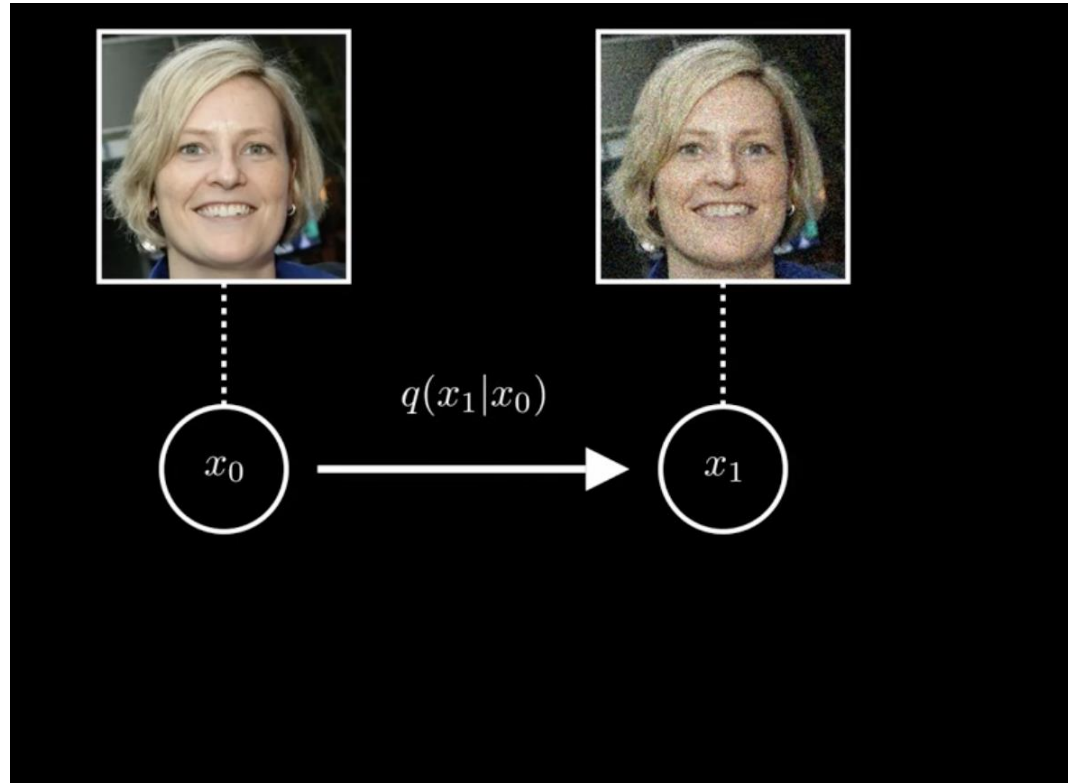
Denoising UNet



x_T

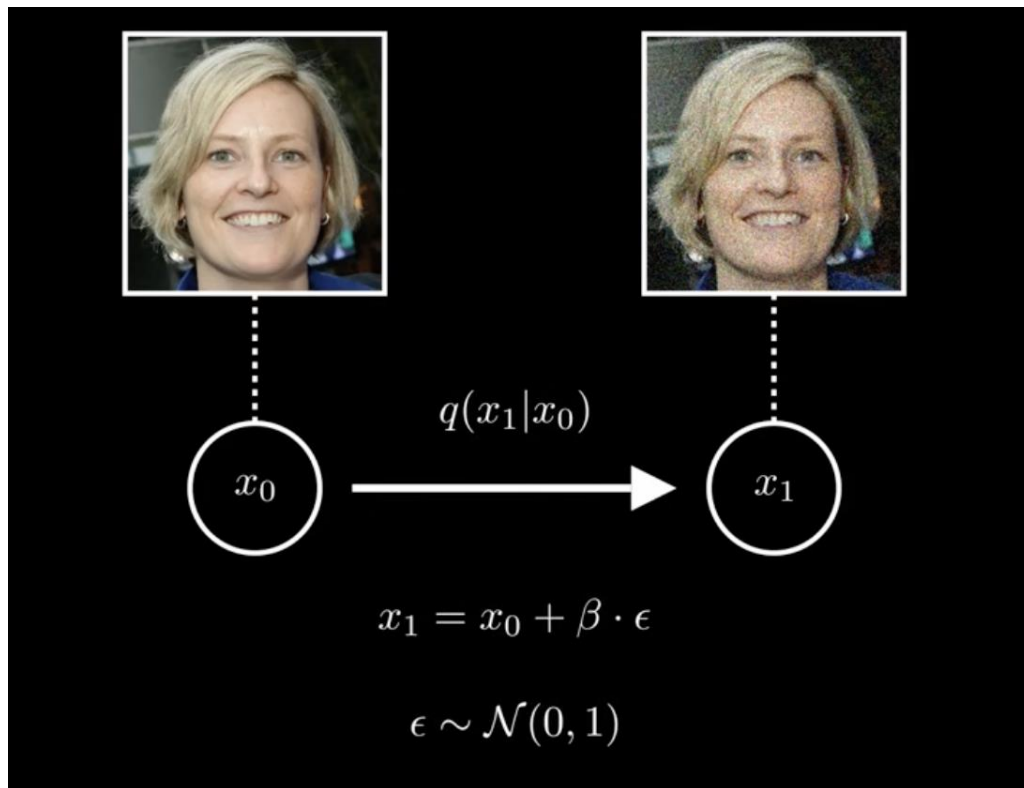
Reverse Diffusion Process

FORWARD DIFFUSION



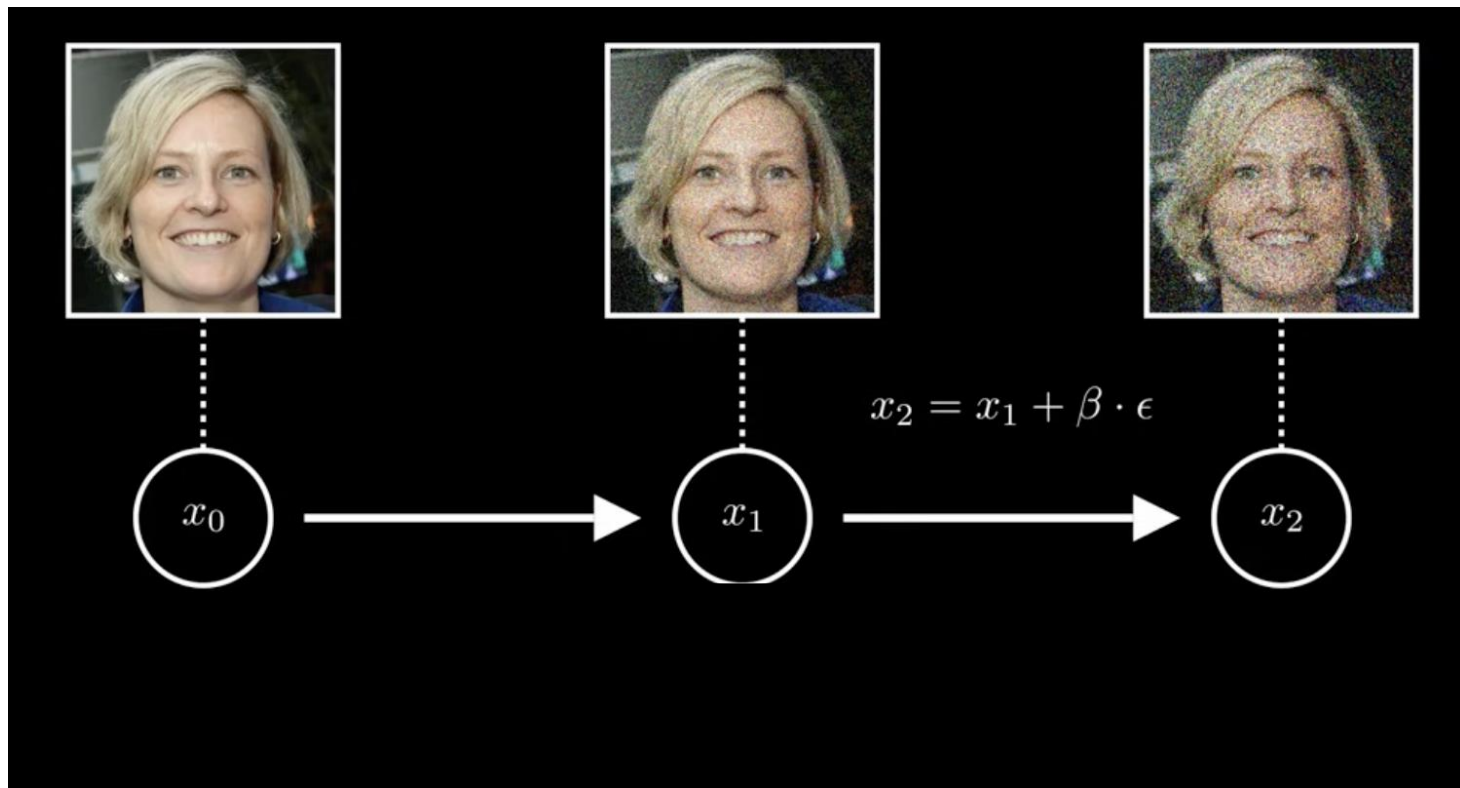
FORWARD DIFFUSION

One step: add a little noise



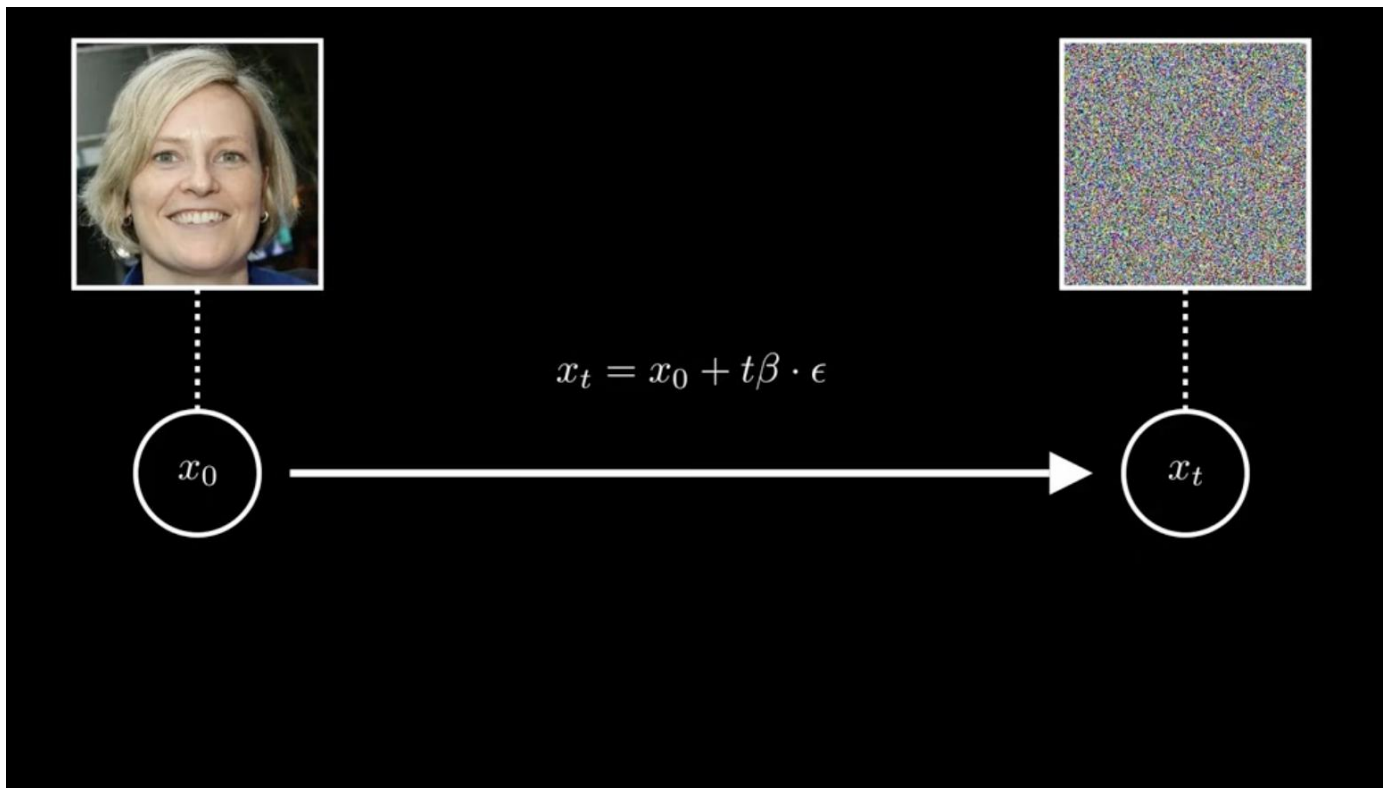
FORWARD DIFFUSION

Every step adds the same β



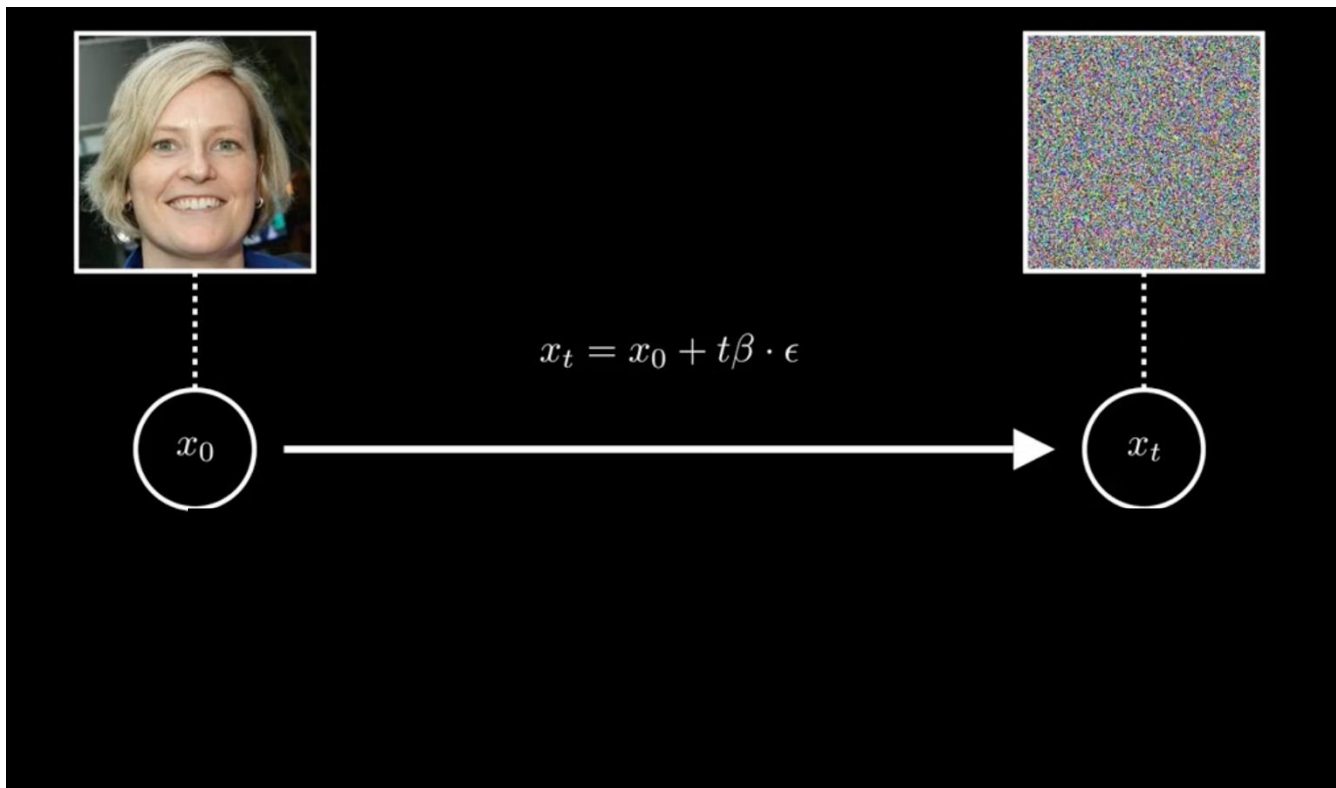
FORWARD DIFFUSION

After t steps



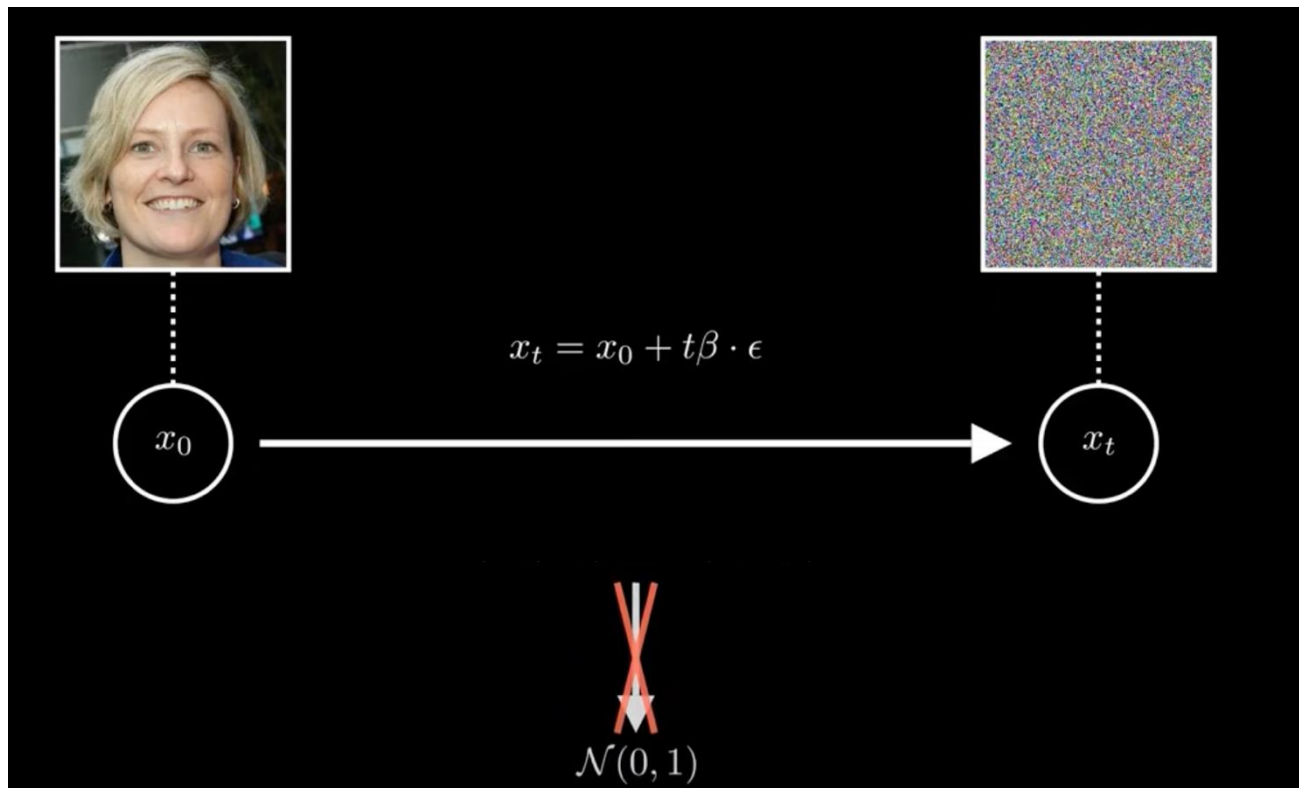
FORWARD DIFFUSION

The distribution after t steps



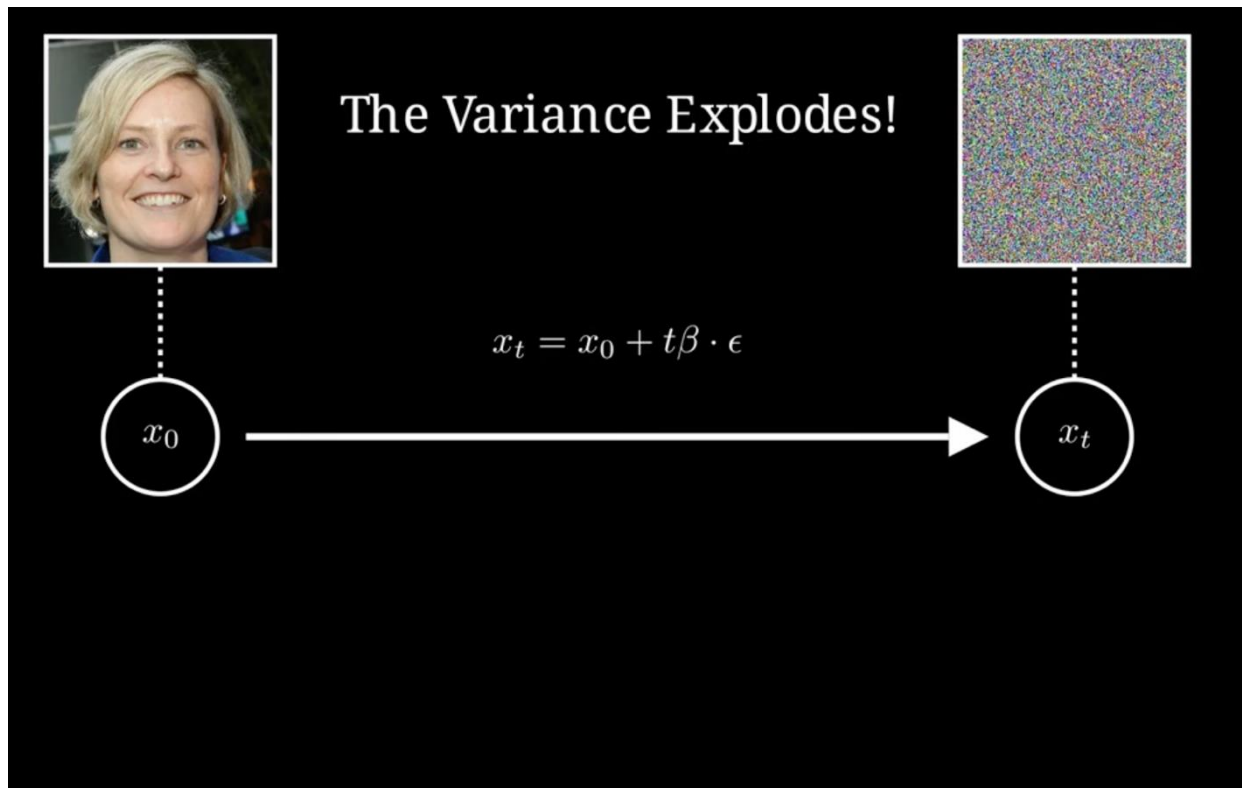
THE PROBLEM

This never reaches $\mathcal{N}(0, 1)$



THE PROBLEM

The variance explodes



THE FIX

What coefficient belongs here?

Objective

$$q(x_t|x_0) \xrightarrow[t \rightarrow \infty]{} \mathcal{N}(0, 1)$$

$$q(x_t|x_{t-1}) = \boxed{?} x_{t-1} + \beta \cdot \epsilon$$

x_{t-1}

x_t

FORWARD DIFFUSION

We want to add Gaussian noise without letting the variance explode.

$$x_t = a x_{t-1} + b \varepsilon$$

$$\varepsilon \sim N(0, 1)$$

So: what should a and b be?

FORWARD DIFFUSION

Take variance on both sides

$$\text{Var}(x_t) = \text{Var}(a x_{t-1}) + \text{Var}(b \varepsilon)$$

THE REQUIREMENT

Keep the variance fixed at 1

If the scale grows at every step, the numbers blow up after a thousand steps.

$$\text{Var}(x_{t-1}) = 1$$

the image, already scaled

$$\text{Var}(\varepsilon) = 1$$

standard normal noise

x_{t-1} and ε are **independent**.

we want $\text{Var}(x_t) = 1$ too

STEP 1

Independent variances add

Scaling by a constant multiplies variance by the constant **squared**.

$$\text{Var}(x_t) = a^2 \text{Var}(x_{t-1}) + b^2 \text{Var}(\varepsilon)$$

with both variances equal to 1:

$$\text{Var}(x_t) = a^2 + b^2$$

STEP 2

Name the noise contribution β

Define β_t to be **the variance the new noise adds** at this step.

$$b^2 = \beta_t$$

take the square root:

$$b = \sqrt{\beta_t}$$

STEP 3

Force the total back to 1

$$a^2 + \beta_t = 1$$

rearrange:

$$a^2 = 1 - \beta_t$$

take the square root:

$$a = \sqrt{1 - \beta_t}$$

THE RESULT

The forward diffusion step

$$x_t = \sqrt{(1 - \beta_t)} x_{t-1} + \sqrt{\beta_t} \varepsilon$$

$$1 - \beta_t$$

the signal's share of the variance

$$\beta_t$$

the noise's share of the variance

The two shares always sum to 1.

SANITY CHECK

Test the two extremes

$$\beta_t = 0$$

$$x_t = x_{t-1}$$

nothing is added

$$0 < \beta_t < 1$$

a mixture

variance stays 1

$$\beta_t = 1$$

$$x_t = \varepsilon$$

the image is gone

The square roots are there because **variances add, not standard deviations.**

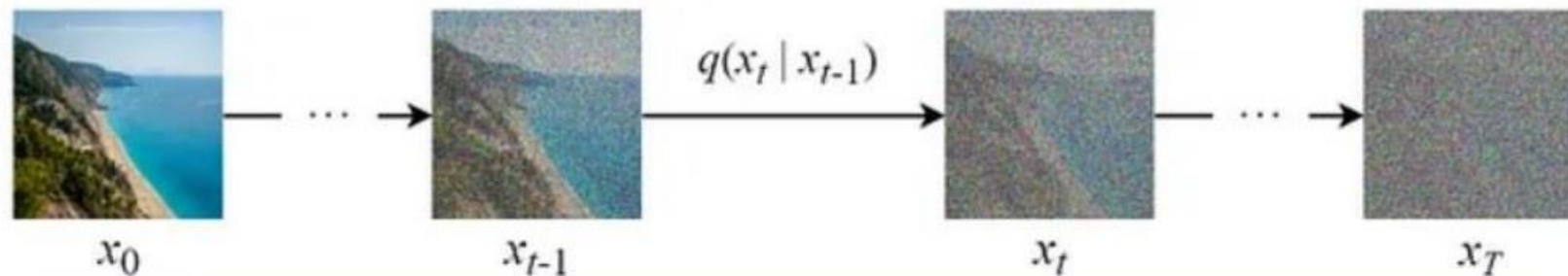
Sample

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1}$$

Distribution

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

Forward Diffusion Process



Distribution of the
noised images

Output

Mean μ_t

Variance Σ_t

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I)$$

Notations:

t : time step (from 1 to T)

x_0 : a data sampled from the real data distribution $q(x)$ (i.e. $x_0 \sim q(x)$)

β_t : variance schedule ($0 \leq \beta_t \leq 1$, and $\beta_0 = \text{small number}$, $\beta_T = \text{large number}$)

I : identity matrix

The forward diffusion process gradually adds Gaussian noise to the input image x_0 step by step, and there will be T steps in total. the process will produce a sequence of noisy image samples $x_1, ..., x_T$.

When $T \rightarrow \infty$, the final result will become a completely noisy image as if it is sampled from an isotropic Gaussian distribution.

But instead of designing an algorithm to iteratively add noise to the image, we can use a closed-form formula to directly sample a noisy image at a specific time step t .

Ancestral Sampling (One Shot)

Ancestral sampling (One Shot)

$$\hat{\alpha}_t = \prod_{i=1}^t \alpha_i$$
$$\alpha_t = 1 - \beta_t$$

$$x_t \sim p(x_t|x_0) = \mathcal{N}\left(x_t; \sqrt{\hat{\alpha}_t} \cdot x_0, (1 - \hat{\alpha}_t)I\right)$$

$$q(x_t|x_0) = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon = \mathcal{N}(x_t, \sqrt{\bar{\alpha}_t}x_0, (1 - \bar{\alpha}_t)I)$$

- Decompose the image generation (sampling) process in many small steps “denosing”
- Small step help to remove noise easily and accurately in backward process
- **Constant Variance:**
- Variance is kept constant, and mean is changed at each time stamp
- Changing both will collapse the structure of the image, and we need to reconstruct from noise.

Closed-Form Formula

The closed-form sampling formula can be derived using the Reparameterization Trick.

If $z \sim \mathcal{N}(\mu, \sigma^2)$ then

$$z = \mu + \sigma \varepsilon \quad \text{where } \varepsilon \sim \mathcal{N}(0, 1)$$

Reparameterization trick

With this trick, we can express the sampled image x_t as follows:

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1}$$

Express x_t using the reparameterization trick

Then we can expand it recursively to get the closed-form formula:

$$\begin{aligned}
 x_t &= \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1} & \varepsilon_0, \dots, \varepsilon_{t-2}, \varepsilon_{t-1} &\sim \mathcal{N}(0, I) \\
 &= \sqrt{\alpha_t} \boxed{x_{t-1}} + \sqrt{1 - \alpha_t} \varepsilon_{t-1} & \bar{\varepsilon}_0, \dots, \bar{\varepsilon}_{t-2}, \bar{\varepsilon}_{t-1} &\sim \mathcal{N}(0, I) \\
 &= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} \varepsilon_{t-2} \right) + \sqrt{1 - \alpha_t} \varepsilon_{t-1} & \varepsilon &\sim \mathcal{N}(0, I) \\
 &= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \boxed{\sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-2}} + \sqrt{1 - \alpha_t} \varepsilon_{t-1} & \alpha_t &= 1 - \beta_t \\
 &= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \boxed{\sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\varepsilon}_{t-2}} & \bar{\alpha}_t &= \prod_{i=1}^t \alpha_i \\
 &\vdots & & \\
 &= \sqrt{\alpha_t \alpha_{t-1} \dots \alpha_1} x_0 + \sqrt{1 - \alpha_t \alpha_{t-1} \dots \alpha_1} \varepsilon \\
 &= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon
 \end{aligned}$$

How?

But how do we jump from line 4 to line 5?

$$\sqrt{\alpha_t(1 - \alpha_{t-1})} \varepsilon_{t-2} + \sqrt{1 - \alpha_t} \varepsilon_{t-1}$$

$$\sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\varepsilon}_{t-2}$$

How?

$$x_t = \sqrt{1 - \beta_t} x_{t-1} + \sqrt{\beta_t} \varepsilon_{t-1}$$

$$\varepsilon_0, \dots, \varepsilon_{t-2}, \varepsilon_{t-1} \sim \mathcal{N}(0, I)$$

$$= \sqrt{\alpha_t} \boxed{x_{t-1}} + \sqrt{1 - \alpha_t} \varepsilon_{t-1}$$

$$\bar{\varepsilon}_0, \dots, \bar{\varepsilon}_{t-2}, \bar{\varepsilon}_{t-1} \sim \mathcal{N}(0, I)$$

$$\varepsilon \sim \mathcal{N}(0, I)$$

$$= \sqrt{\alpha_t} \left(\sqrt{\alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_{t-1}} \varepsilon_{t-2} \right) + \sqrt{1 - \alpha_t} \varepsilon_{t-1}$$

$$\alpha_t = 1 - \beta_t$$

$$= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \boxed{\sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-2}} + \boxed{\sqrt{1 - \alpha_t} \varepsilon_{t-1}}$$

$$\bar{\alpha}_t = \prod_{i=1}^t \alpha_i$$

$X + Y$

Reparameterization Trick

Underlying Normal Distribution

$$0 + \sqrt{\alpha_t (1 - \alpha_{t-1})} \varepsilon_{t-2} \quad \text{-----} \quad X \sim \mathcal{N}(0, \alpha_t (1 - \alpha_{t-1}) I)$$

$$0 + \sqrt{1 - \alpha_t} \varepsilon_{t-1} \quad \text{-----} \quad Y \sim \mathcal{N}(0, (1 - \alpha_t) I)$$

Recall:

$$\text{If } X \sim \mathcal{N}(\mu_X, \sigma_X^2) \quad Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$$

$$Z = X + Y$$

$$\text{Then } Z \sim \mathcal{N}(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$$

$$Z \sim \mathcal{N}(0, \sigma_X^2 + \sigma_Y^2)$$

$$Z \sim \mathcal{N}(0, (1 - \alpha_t \alpha_{t-1}) I)$$

Reparameterization Trick
↓

$$0 + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\epsilon}_{t-2}$$

$$\begin{aligned} \sigma_X^2 + \sigma_Y^2 &= \alpha_t(1 - \alpha_{t-1}) + (1 - \alpha_t) \\ &= \cancel{\alpha_t} - \alpha_t \alpha_{t-1} + 1 - \cancel{\alpha_t} \\ &= 1 - \alpha_t \alpha_{t-1} \end{aligned}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} x_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\epsilon}_{t-2}$$

⋮

$$= \sqrt{\alpha_t \alpha_{t-1} \dots \alpha_1} x_0 + \sqrt{1 - \alpha_t \alpha_{t-1} \dots \alpha_1} \epsilon$$

$$= \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

Detailed derivation from line 4 to line 5

Repeating these steps will give us the following formula which depends only on the input image x_0 :

$$x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

The closed-form formula

Now we can directly sample x_t at any time step using this formula, and this makes the forward process much faster.

where:

To extract ϵ_t (the noise at step t) from a given noisy sample x_t :

$$\epsilon_t = \frac{x_t - \sqrt{\bar{\alpha}_t} x_0}{\sqrt{1 - \bar{\alpha}_t}}$$

- x_0 is the original clean image,
- x_t is the noisy image at timestep t ,
- $\bar{\alpha}_t$ is the accumulated product of all previous noise scales,
- $\epsilon \sim \mathcal{N}(0, I)$ is standard Gaussian noise.

How Do We Compute $\bar{\alpha}_t$?

$\bar{\alpha}_t$ is the cumulative product of a sequence of predefined noise scales α_t . These α_t values are usually determined by a noise schedule, which is a function that controls how noise is added across timesteps. The most common schedules include:

- **Linear schedule:** β_t increases linearly from a small to a large value.
- **Cosine schedule:** Inspired by cosine functions for smoother noise distribution.

Once we have a noise schedule, we compute:

$$\alpha_t = 1 - \beta_t$$

$$\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$$

the noise schedule provides all the α_t values upfront, allowing you to calculate any $\bar{\alpha}_t$ directly without needing to step through the diffusion process.

$\bar{\alpha}_t$: precomputed from schedule

This cumulative product accumulates the effect of all past timesteps.

If you only have the original image x_0 , you must define a noise schedule first (e.g., a linear or cosine schedule). Then, compute $\bar{\alpha}_t$ as the cumulative product of α_t up to timestep t . This allows you to generate noisy samples in one step using the equation.

2 How to Compute Noisy Image for Any Timestamp

Here's the magic: **closed-form formula**

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$$

Where:

- x_0 : clean image
- t : chosen timestep
- $\bar{\alpha}_t$: precomputed from schedule
- $\varepsilon \sim \mathcal{N}(0, I)$: fresh Gaussian noise

So even if you want **step 100**, you just:

1. Look up $\bar{\alpha}_{100}$ from your precomputed array
2. Sample Gaussian noise ε
3. Compute x_{100} using the above equation

You **don't need to iterate step-by-step** ($1 \rightarrow 2 \rightarrow 3 \dots \rightarrow 100$), which would be very slow.

Symbol	Meaning	Computation
β_t	Variance of noise at step t	from schedule (linear/cosine)
α_t	Signal retained per step	$1 - \beta_t$
$\bar{\alpha}_t$	Cumulative signal left after t steps	product of $\alpha_1 \dots \alpha_t$
x_t	Noisy image at step t	$\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$